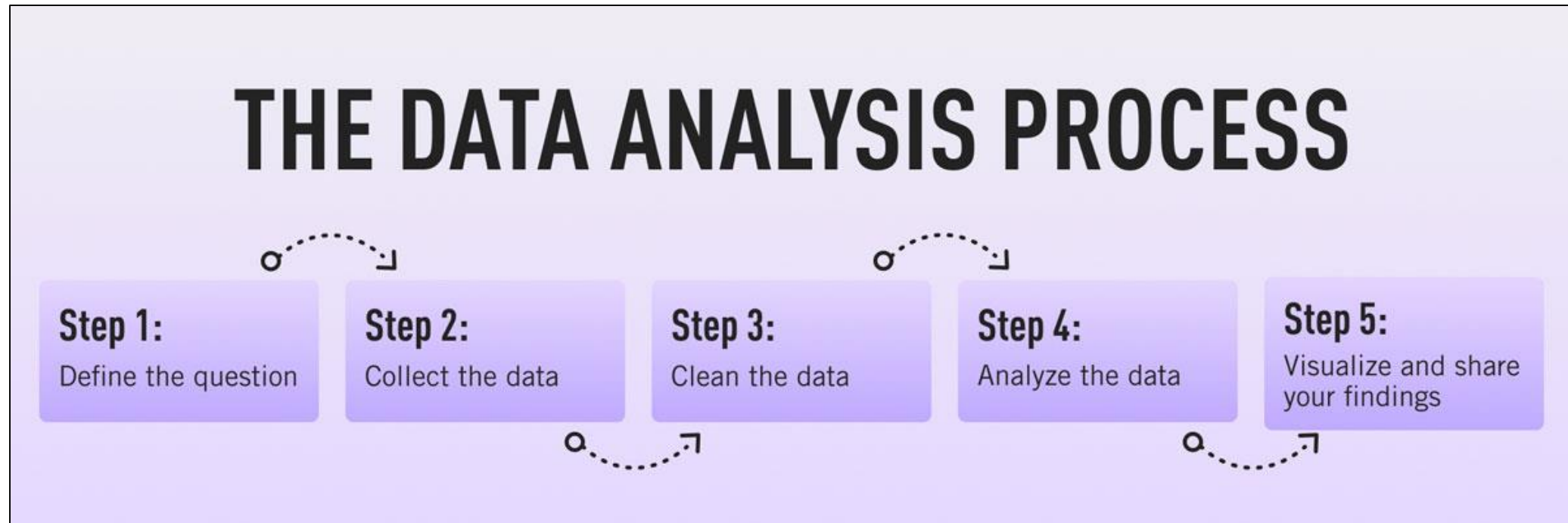


Explain the step-by-step process that a data analyst typically follows in their workflow.

- ❖ Like any scientific discipline, data analysis follows a rigorous step-by-step process. Each stage requires different skills and know-how.
- ❖ To get meaningful insights, though, it's important to understand the process as a whole. An underlying framework is invaluable for producing results that stand up to scrutiny.



Step 1: Define the question(s) you want to answer

- ❖ In the first step of process the data analyst is given a problem/business task. The analyst has to understand the task and the stakeholder's expectations for the solution.
- ❖ A stakeholder is a person that has invested their money and resources to a project. The analyst must be able to ask **different questions in order to find the right solution to their problem.**
- ❖ The analyst has to find the root cause of the problem in order to fully understand the problem.
- ❖ The analyst must make sure that he/she doesn't have any distractions while analyzing the problem.
- ❖ Communicate effectively with the stakeholders and other colleagues to completely understand what the underlying problem is.
- ❖ At this stage, you'll take a clearly defined problem and come up with a relevant question or hypothesis you can test.
- ❖ **Questions to ask yourself for the Ask phase are:**
 - ❖ What are the problems that are being mentioned by my stakeholders?
 - ❖ What are their expectations for the solutions?

Step 2: Collect the data

- ❖ The second step is to Prepare or Collect the Data. This step includes collecting data and storing it for further analysis.
- ❖ The analyst has to collect the data based on the task given from multiple sources. The data has to be collected from various sources, internal or external sources.
- ❖ Internal data is the data available in the organization that you work for while external data is the data available in sources other than your organization.
- ❖ The data that is collected by an individual from their own resources is called first-party data.
- ❖ The data that is collected and sold is called second-party data.
- ❖ Data that is collected from outside sources is called third-party data. The common sources from where the data is collected are Interviews, Surveys, Feedback, Questionnaires. The collected data can be stored in a spreadsheet or SQL database.
- ❖ Data analysts will usually gather structured data from primary or internal sources, such as CRM software or email marketing tools.

Step 3: Clean the data

- ❖ The third step is Clean and Process Data. After the data is collected from multiple sources, it is time to clean the data.
- ❖ Clean data means data that is free from **misspellings, redundancies, and irrelevance**.
- ❖ Clean data largely depends on data integrity. There might be duplicate data or the data might not be in a format, therefore the unnecessary data is removed and cleaned.
- ❖ There are different functions provided by SQL and Excel to clean the data. **This is one of the most important steps in Data Analysis as clean and formatted data helps in finding trends and solutions.**
- ❖ The most important part of the Process phase is to check whether your data is biased or not. Bias is an act of favoring a particular group/community while ignoring the rest. Biasing is a big no-no as it might affect the overall data analysis. The data analyst must make sure to include every group while the data is being collected.
- ❖ Your original dataset may contain duplicates, anomalies, or missing data which could distort how the data is interpreted, so these all need to be removed.
- ❖ Data cleaning can be a time-consuming task, but it's crucial for obtaining accurate results.
- ❖ Ex. Field extraction algorithm, Clustering .

Step 4: Analyze the data

- ❖ The fourth step is to Analyze. The cleaned data is used for analyzing and identifying trends.
- ❖ It also performs calculations and combines data for better results. The tools used for performing calculations are Excel or SQL.
- ❖ These tools provide in-built functions to perform calculations or sample code is written in SQL to perform calculations.
- ❖ Using Excel, we can create pivot tables and perform calculations while SQL creates temporary tables to perform calculations.
- ❖ Programming languages are another way of solving problems. They make it much easier to solve problems by providing packages. The most widely used programming languages for data analysis are R and Python.
- ❖ Some common techniques include **regression analysis, cluster analysis, and time-series analysis.**
- ❖ **This step in the process also ties in with the four different types of analysis we looked at in (Descriptive, Diagnostic, Predictive, and Prescriptive).**

Step 5: Interpret and share the results

- ❖ This final step in the process is where data is transformed into valuable business insights.
- ❖ Depending on the type of analysis conducted, you'll present your findings in a way that others can understand—in the form of a chart or graph.
- ❖ The data now transformed has to be made into a visual (chart, graph).
- ❖ The reason for making data visualizations is that there might be people, mostly stakeholders that are non-technical.
- ❖ Visualizations are made for a simple understanding of complex data. **Tableau** and **Looker** are the two popular tools used for compelling data visualizations.
- ❖ **Tableau** is a simple drag and drop tool that helps in creating compelling visualizations.
- ❖ **Looker is a data viz tool** that directly connects to the database and creates visualizations.
- ❖ Tableau and Looker are both equally used by data analysts for creating a visualization.
- ❖ R and Python have some packages that provide beautiful data visualizations.
- ❖ R has a package named **ggplot** which has a variety of data visualizations.
- ❖ A presentation is given based on the data findings.